

use and work on all of them. The ultimate users should not suffer any major consequences due to shortcomings in the app. A mobile app developer not only takes care of the development of the mobile application but also needs to address aspects like usability, performance, efficiency and other such user expectations. In order to incorporate all these aspects without any bugs, a mobile application development framework is used, which brings different mobile application developers onto the same platform.

Mobile app development for different OSs

Developing apps for mobile devices requires taking into consideration the different constraints and features of these devices. These include a wide array of specifications such as screen size, hardware specs, mobile configurations and the platform or OS used by the mobile device. And because operating systems are so different, apps developed for iOS, for instance, may not work on Android.

Mobile application development for Android: As of 2016, Android had the largest installed base of any mobile operating system, and the most number of mobile apps built so far are for Android. The Android OS uses a Linux kernel with some higher-level APIs, which are written in C. Different mobile apps running on this OS are normally programmed in Java, and they run with the Dalvik virtual machine using so called just-in-time compilation to translate the Java byte code into Dalvik dex-code. This adds some of the secure features to these apps, like efficient shared memory management, UNIX user identifiers (UIDs), pre-emptive multi-tasking, and file permission with the type safe concept of Java. Android Studio is widely used to develop different mobile apps.

Mobile application development for iOS: iOS is a mobile operating system developed by Apple Inc. and is used exclusively for Apple hardware. It's built on the open source Darwin core operating system. Most of the apps in Apple's App store are written using the Objective-C programming language. iOS mobile app developers typically use Xcode. To build an iOS app, a developer must use Mac OS X since other operating systems are not supported. The development tools that are required for this are iOS 7 SDK and Xcode 5. The built iOS mobile app can be run using the iOS simulator, which is a part of the iOS SDK. In order to run these apps on a real device, they have to be made available in Apple's App store.

Mobile application development for Windows: Windows Phone is a closed source smartphone operating system developed by Microsoft. The Windows Phone kernel handles access to the low-level device drivers. It also takes care of basic security, storage and the networking features of the mobile phone. Any Windows mobile app can make use of these features. Apart from these, there are three libraries - App Model for application



Figure 1: Categories of mobile applications for development
(Image source: googleimages.com)



Figure 2: Some important facts for mobile applications and users
(Image source: googleimages.com)

management, a cloud integration module for Web searches via Bing, and a UI model for user interface management. All three of these can be used to develop apps for Windows. XAML is used by Windows Phone apps for the UI and C# or Visual Basic for the code development. Those familiar with C++ can develop a Direct3D app for Windows Phone 8 in that language by taking full advantage of the phone's graphics hardware.

Cross-platform technology of mobile app development

The mobile apps developed using cross-platform or hybrid technology can run on more than one platform at a time, without any intervention from developers to create separate versions of application code. Hybrid mobile applications are essentially written as Web applications using HTML, JavaScript and CSS. They are already embedded within a 'native wrapper' allowing different applications built using